

Introductory Statistics for Econometrics

Unit 8. Point Estimation:
MLE and OLS

Estimators and their sampling distribution; bias, consistency, efficiency; the simple linear regression model and OLS

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Learning outcomes. After this unit the student should be able to (i) define an estimator and its sampling distribution; (ii) check whether an estimator is unbiased, consistent, and efficient; (iii) construct estimators by Maximum Likelihood; (iv) state the Stock–Watson Least-Squares Assumptions for the simple linear regression model; (v) derive the OLS estimator and its sampling properties (unbiasedness, consistency, BLUE, asymptotic normality); (vi) recognise OLS as Maximum Likelihood estimator under suitable assumptions.

Source material. Condensed from MIT OCW 14.30 (Spring 2009), K. Menzel, Lecture Notes 17–18 for the general estimation theory. The OLS section follows the framing of Stock and Watson, *Introduction to Econometrics*, Chapters 3–5.

1 Estimators and their sampling distribution

We observe an i.i.d. random sample $X_1, \dots, X_n$ drawn from a parametric family $\{f(x | \theta) : \theta \in \Theta\}$. The true parameter θ_0 is unknown. The goal of *point estimation* is to use the sample to produce a single value $\hat{\theta} \in \Theta$ that is, in some sense, close to θ_0 .

Definition 1.1 (Statistic, estimator, estimate). A *statistic* is any function of the sample, $T = T(X_1, \dots, X_n)$. An *estimator* of θ is a statistic $\hat{\theta} = \hat{\theta}(X_1, \dots, X_n)$ targeting θ_0 . The realised value $\hat{\theta}(x_1, \dots, x_n)$ for a particular sample is the corresponding *estimate*.

Because $\hat{\theta}$ is a function of random variables, it is itself random. Its distribution is called the *sampling distribution* of $\hat{\theta}$ — the central object of all statistical inference. We will study three basic features of this distribution: bias, consistency, and variance.

Why three properties? An estimator can hit the target on average but be very noisy (low bias, high variance) or be very stable but biased (low variance, high bias). The trade-off is captured by the *mean squared error*

$$\text{MSE}(\hat{\theta}) = \mathbb{E}_{\theta_0}[(\hat{\theta} - \theta_0)^2] = \text{Bias}(\hat{\theta})^2 + \text{Var}(\hat{\theta}),$$

which we will return to repeatedly.

2 Bias, consistency, efficiency

Definition 2.1 (Bias and unbiasedness). The *bias* of $\hat{\theta}$ is $\text{Bias}(\hat{\theta}) = \mathbb{E}_{\theta_0}[\hat{\theta}] - \theta_0$. The estimator is *unbiased* if $\mathbb{E}_{\theta_0}[\hat{\theta}] = \theta_0$ for every θ_0 .

Definition 2.2 (Consistency). $\hat{\theta}$ is *consistent* for θ if $\hat{\theta} \xrightarrow{p} \theta_0$ as $n \rightarrow \infty$, for every θ_0 .

Bias and consistency are independent properties.

- *Unbiased but not consistent.* $\tilde{\theta}(X_1, \dots, X_n) = X_n$ (last observation only) is unbiased for μ when $\mathbb{E}[X_i] = \mu$, but its variance does not vanish, so it is not consistent.
- *Biased but consistent.* For $X_i \sim U[0, \theta_0]$, $\hat{\theta}_1 = \max\{X_1, \dots, X_n\}$ has $\mathbb{E}[\hat{\theta}_1] = n\theta_0/(n+1) < \theta_0$ (biased downward), but $\hat{\theta}_1 \xrightarrow{p} \theta_0$ (consistent).

Definition 2.3 (Standard error and efficiency). The *standard error* of $\hat{\theta}$ is $\text{SE}(\hat{\theta}) = \sqrt{\text{Var}(\hat{\theta})}$ (or its estimated counterpart $\widehat{\text{SE}}$). Among unbiased estimators of θ , $\hat{\theta}_A$ is *more efficient than* $\hat{\theta}_B$ if $\text{Var}(\hat{\theta}_A) \leq \text{Var}(\hat{\theta}_B)$ for every θ_0 . An estimator is *efficient* in a class if it has the smallest variance in that class.

Example 2.4 (Efficient combination of two unbiased estimators). Two test scores X, Y have $\mathbb{E}[X] = \mathbb{E}[Y] = \mu$, variances σ_X^2, σ_Y^2 and covariance σ_{XY} . Among all weighted averages $pX + (1-p)Y$ (each unbiased), the variance-minimising weight is

$$p^* = \frac{\sigma_Y^2 - \sigma_{XY}}{\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}} = \frac{\text{Cov}(Y - X, Y)}{\text{Var}(Y - X)}.$$

This weight is the OLS coefficient from regressing Y on $Y - X$ — the same ratio of covariance to variance that we will see in Section 6.

3 Sample mean and sample variance

Sample mean. For an i.i.d. sample with $\mathbb{E}[X_i] = \mu$ and $\text{Var}(X_i) = \sigma^2$,

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i, \quad \mathbb{E}[\bar{X}_n] = \mu, \quad \text{Var}(\bar{X}_n) = \frac{\sigma^2}{n}.$$

By the LLN $\bar{X}_n \xrightarrow{p} \mu$ (consistent) and by the CLT $\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$ (asymptotically normal). The sample mean is unbiased, consistent and, in the class of *linear* unbiased estimators of μ , efficient.

Sample variance. The natural plug-in estimator $\hat{\sigma}^2 = \frac{1}{n} \sum (X_i - \bar{X}_n)^2$ is biased:

$$\mathbb{E}[\hat{\sigma}^2] = \frac{n-1}{n} \sigma^2.$$

Estimating the mean “costs one degree of freedom”. The corrected estimator

$$S_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2 \tag{1}$$

is unbiased, $\mathbb{E}[S_n^2] = \sigma^2$, and consistent. The standard error of the sample mean is then estimated by $\widehat{\text{SE}}(\bar{X}_n) = S_n/\sqrt{n}$.

4 Maximum Likelihood Estimation

The *Maximum Likelihood* principle uses all the information in the density, not just selected moments.

Definition 4.1 (Likelihood and MLE). For an i.i.d. sample with density $f(\cdot | \theta)$, the *likelihood* is

$$\mathcal{L}_n(\theta) = \prod_{i=1}^n f(X_i | \theta), \quad \ell_n(\theta) = \log \mathcal{L}_n(\theta) = \sum_{i=1}^n \log f(X_i | \theta).$$

The *Maximum Likelihood Estimator* (MLE) is

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta \in \Theta} L_n(\theta).$$

In other words, the MLE chooses the parameter value that assigns the highest likelihood to the observed sample $\{X_1, \dots, X_n\}$.

Example 4.2. Suppose there are two possible observed samples, $x^{(1)}$ and $x^{(2)}$, and two possible parameter values, θ_1 and θ_2 . The likelihood values are: If we observe the sample $x^{(1)}$, then

Table 1: Likelihood values $L_n(\theta)$

$L_n(\theta)$	$x^{(1)}$	$x^{(2)}$
θ_1	$\frac{1}{4}$	$\frac{1}{8}$
θ_2	$\frac{1}{8}$	$\frac{1}{4}$

$$\hat{\theta}_{\text{MLE}} = \theta_1,$$

because $L_n(\theta_1) > L_n(\theta_2)$ for the observed sample $x^{(1)}$. Notice that the MLE uses the observed sample to construct the likelihood function. Hence, different samples can lead to different likelihood functions and, consequently, to different MLEs.

Example 4.3 (Normal MLE). For $X_i \sim \mathcal{N}(\mu, \sigma^2)$,

$$\ell_n(\mu, \sigma^2) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu)^2.$$

First-order conditions give

$$\hat{\mu}_{\text{MLE}} = \bar{X}_n, \quad \widehat{\sigma^2}_{\text{MLE}} = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2.$$

Note: the MLE of σ^2 is the *biased* estimator from Section 3; it is consistent but not unbiased.

Example 4.4 (Uniform MLE on a positive interval). For $X_i \sim U[0, \theta]$, $\mathcal{L}_n(\theta) = \theta^{-n} \mathbf{1}\{\theta \geq \max_i X_i\}$, so

$$\hat{\theta}_{\text{MLE}} = \max\{X_1, \dots, X_n\}.$$

Since $\mathbb{E}[\hat{\theta}_{\text{MLE}}] = n\theta_0/(n+1) < \theta_0$, the MLE is biased but consistent. Comparison with $\hat{\theta}_{\text{MoM}} = 2\bar{X}_n$ (unbiased): the MLE has lower MSE for moderate n — a typical pattern.

Properties of the MLE. Under regularity (smooth density, identifiable θ , finite Fisher information),

- *Consistent*: $\hat{\theta}_{\text{MLE}} \xrightarrow{p} \theta_0$.
- *Asymptotically normal*: $\sqrt{n}(\hat{\theta}_{\text{MLE}} - \theta_0) \xrightarrow{d} \mathcal{N}(0, I(\theta_0)^{-1})$, where $I(\theta_0) = -\mathbb{E}[\partial^2 \log f / \partial \theta^2]$ is the Fisher information.
- *Asymptotically efficient*: among consistent estimators, MLE attains the Cramér–Rao lower bound $I(\theta_0)^{-1}/n$.

Caveats. MLE may be biased in finite samples, can be hard to compute, and is sensitive to misspecification of f .

5 The simple linear regression model

The remainder of the unit specialises to the central model of econometrics following the framing of Stock & Watson, *Introduction to Econometrics*, Chapters 3–5.

We observe a random sample $\{(X_i, Y_i)\}_{i=1}^n$. The *population regression function* is the conditional expectation $\mu(x) = \mathbb{E}[Y \mid X = x]$ from Unit 6. We *model* this conditional expectation as linear:

$$Y_i = \beta_0 + \beta_1 X_i + u_i, \quad i = 1, \dots, n, \quad (2)$$

or in matrix form,

$$Y_i = \beta' \mathbf{X}_i + u_i, \quad i = 1, \dots, n, \quad (3)$$

where $\beta = (\beta_0, \beta_1)'$, $\mathbf{X}_i = (1, X_i)$ and u_i is an unobserved *error term*. The intercept β_0 and slope β_1 are population parameters and the targets of estimation.

The Stock–Watson Least-Squares Assumptions.

(LSA1) *Conditional mean zero (exogeneity):* $\mathbb{E}[u_i \mid X_i] = 0$. This implies $\mathbb{E}[u_i] = 0$ and $\text{Cov}(X_i, u_i) = 0$. Under (LSA1) the model coincides with the conditional expectation function: $\mathbb{E}[Y \mid X = x] = \beta_0 + \beta_1 x$.

(LSA2) *i.i.d. sampling:* (X_i, Y_i) are i.i.d. across i .

(LSA3) *Large outliers are unlikely:* $\mathbb{E}[X^4], \mathbb{E}[Y^4] < \infty$ (finite fourth moments).

These are the assumptions under which OLS will be unbiased, consistent and asymptotically normal. Two optional extras are useful in addition:

(LSA4) *Homoskedasticity:* $\text{Var}(u_i \mid X_i) = \sigma_u^2$, a constant not depending on X . Under homoskedasticity OLS achieves the Gauss–Markov bound (BLUE).

(LSA5) *Conditional normality:* $u_i \mid X_i \sim \mathcal{N}(0, \sigma_u^2)$. Under (LSA1)–(LSA5) OLS coincides with the MLE.

6 The OLS estimator

Following the population logic of Unit 6, OLS chooses $(\hat{\beta}_0, \hat{\beta}_1)$ to minimise the in-sample mean squared error,

$$(\hat{\beta}_0, \hat{\beta}_1) = \arg \min_{b_0, b_1} \sum_{i=1}^n (Y_i - b_0 - b_1 X_i)^2. \quad (4)$$

Setting the partial derivatives to zero gives the *normal equations*

$$\sum_i (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = 0, \quad \sum_i X_i (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = 0.$$

Solving:

$$\boxed{\hat{\beta}_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{s_{XY}}{s_X^2}, \quad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}.} \quad (5)$$

The OLS slope is the sample analogue of the population OLS coefficient $\text{Cov}(X, Y)/\text{Var}(X)$ derived in Unit 6.

Algebraic properties of OLS. Define the fitted values $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$ and residuals $\hat{u}_i = Y_i - \hat{Y}_i$. Then mechanically:

- (i) $\sum_i \hat{u}_i = 0$;
- (ii) $\sum_i X_i \hat{u}_i = 0$ (residuals orthogonal to the regressor);
- (iii) $\bar{Y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{X}$ (the OLS line passes through the sample mean);
- (iv) $\text{TSS} = \text{ESS} + \text{RSS}$, where $\text{TSS} = \sum (Y_i - \bar{Y})^2$, $\text{ESS} = \sum (\hat{Y}_i - \bar{Y})^2$, $\text{RSS} = \sum \hat{u}_i^2$. The coefficient of determination is $R^2 = \text{ESS}/\text{TSS} = 1 - \text{RSS}/\text{TSS}$.

7 Sampling properties of OLS

Theorem 7.1 (Unbiasedness and consistency under LSA1–LSA3). *Under (LSA1)–(LSA3) and $\text{Var}(X) > 0$,*

- (i) $\mathbb{E}[\hat{\beta}_1 | X_1, \dots, X_n] = \beta_1$; hence $\mathbb{E}[\hat{\beta}_1] = \beta_1$ (*unbiasedness*).
- (ii) $\hat{\beta}_1 \xrightarrow{p} \beta_1$ as $n \rightarrow \infty$ (*consistency*).
- (iii) *Asymptotic normality:* $\sqrt{n}(\hat{\beta}_1 - \beta_1) \xrightarrow{d} \mathcal{N}\left(0, \frac{\text{Var}[(X_i - \mu_X)u_i]}{[\text{Var}(X_i)]^2}\right)$.

Identical statements hold for $\hat{\beta}_0$.

Heteroskedasticity-robust standard errors. The general form of the asymptotic variance does not assume homoskedasticity. Estimating it gives the *heteroskedasticity-robust* (or “White”) standard errors,

$$\widehat{\text{SE}}(\hat{\beta}_1)^2 = \frac{1}{n} \frac{\frac{1}{n-2} \sum_i (X_i - \bar{X})^2 \hat{u}_i^2}{\left[\frac{1}{n} \sum_i (X_i - \bar{X})^2\right]^2}.$$

This is the standard error reported by Stata’s `regress`, `robust` and R’s `coeftest(..., vcov = vcovHC)`. Use it by default in empirical work.

Under (LSA4): Gauss–Markov.

Theorem 7.2 (Gauss–Markov). *Under (LSA1)–(LSA4), OLS is the best linear unbiased estimator (BLUE) of (β_0, β_1) : among all linear unbiased estimators, it has the smallest variance. Moreover, the conditional variance simplifies to*

$$\text{Var}(\hat{\beta}_1 | X_1, \dots, X_n) = \frac{\sigma_u^2}{\sum_{i=1}^n (X_i - \bar{X})^2}.$$

The corresponding *homoskedasticity-only* standard error is $\widehat{\text{SE}}_{\text{ho}}(\hat{\beta}_1) = s_u / \sqrt{\sum (X_i - \bar{X})^2}$, with $s_u^2 = \text{RSS}/(n-2)$.

Remark 7.3 (When to use which standard error). Heteroskedasticity-robust SEs are valid under (LSA1)–(LSA3) regardless of homoskedasticity; homoskedasticity-only SEs are smaller *when* homoskedasticity holds, but invalid otherwise. Modern empirical practice defaults to robust standard errors.

8 OLS as MLE

The two principles in this unit converge on the same OLS coefficient.

OLS as MLE under conditional normality. Under (LSA1)–(LSA5), the conditional density of $Y_i \mid X_i$ is $\mathcal{N}(\beta_0 + \beta_1 X_i, \sigma_u^2)$ and

$$\ell_n(\beta_0, \beta_1, \sigma_u^2) = -\frac{n}{2} \log(2\pi\sigma_u^2) - \frac{1}{2\sigma_u^2} \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i)^2.$$

Maximising over (β_0, β_1) is equivalent to *minimising* the residual sum of squares — the OLS criterion. Hence under conditional normality, OLS coincides with the MLE.

Example 8.1 (Putting it all together). A student tracks her commute time Y_i on day i and the number of buses she takes $X_i \in \{0, 1, 2\}$. With 60 days of data she computes $\bar{X} = 0.95$, $\bar{Y} = 28$ minutes, $s_X^2 = 0.45$, $s_{XY} = 2.7$. Then $\hat{\beta}_1 = 2.7/0.45 = 6$ minutes per bus and $\hat{\beta}_0 = 28 - 6 \cdot 0.95 = 22.3$. Under (LSA1)–(LSA3) these are consistent estimates of the conditional expectation $\mathbb{E}[\text{commute} \mid \text{buses} = x] = 22.3 + 6x$. Inference on the slope (e.g. a 95% confidence interval) is the subject of Unit 10.

Summary

- An estimator $\hat{\theta}$ is a function of the data; its sampling distribution drives all inference.
- Three properties: *unbiased* ($\mathbb{E}[\hat{\theta}] = \theta_0$), *consistent* ($\hat{\theta} \xrightarrow{p} \theta_0$), *efficient* (smallest variance among unbiased competitors). $\text{MSE} = \text{bias}^2 + \text{variance}$.
- Sample mean $\bar{X}_n$: unbiased, consistent, BLUE for μ in the class of linear estimators. Sample variance S_n^2 uses $n - 1$ in the denominator to remove the bias.
- *Maximum Likelihood*: maximise $\ell_n(\theta)$. Consistent, asymptotically normal with variance $I(\theta_0)^{-1}/n$, asymptotically efficient. May be biased in finite samples.
- *Linear regression model* (Stock–Watson): $Y = \beta_0 + \beta_1 X + u$, with assumptions (LSA1) $\mathbb{E}[u \mid X] = 0$, (LSA2) i.i.d., (LSA3) finite fourth moments.
- *OLS estimator*: $\hat{\beta}_1 = s_{XY}/s_X^2$, $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$. Unbiased, consistent, and asymptotically normal under (LSA1–3); BLUE under added homoskedasticity (LSA4); MLE under added conditional normality (LSA5).
- OLS is the conditional MLE under normality.

Suggested exercises

1. Show that for an i.i.d. sample with $\mathbb{E}[X_i] = \mu$ and $\text{Var}(X_i) = \sigma^2 < \infty$, the sample variance S_n^2 in (1) is unbiased for σ^2 .
2. For $X_i \sim \text{Exp}(\lambda)$, derive MLE estimator of λ . Is it unbiased? Consistent?
3. Verify items (i)–(iii) of the algebraic OLS properties in Section 6 from the normal equations.
4. Show that under (LSA1)–(LSA3), $\hat{\beta}_1$ in (5) is conditionally unbiased: write it as $\beta_1 + \sum_i w_i u_i$ with $w_i = (X_i - \bar{X}) / \sum (X_j - \bar{X})^2$ and apply (LSA1).
5. (*Stock–Watson Ch. 4 style.*) Generate $X \sim U[0, 1]$ and $u \sim \mathcal{N}(0, 1)$ independently, set $Y = 1 + 2X + u$. Simulate 1,000 samples of size $n = 200$, compute $\hat{\beta}_1$ in each, and verify that the empirical distribution of $\sqrt{n}(\hat{\beta}_1 - 2)$ is approximately normal.
6. Consider OLS in matrix form: $\hat{\beta} = (X^\top X)^{-1} X^\top \mathbf{Y}$. Show that under the LSAs (multivariate version), $\hat{\beta}$ is unbiased and asymptotically MVN with covariance $\sigma_u^2 (X^\top X)^{-1}$ (homoskedastic) or its robust analogue.

Material adapted from MIT OpenCourseWare 14.30 *Introduction to Statistical Methods in Economics* (Spring 2009), Konrad Menzel, Lecture Notes 17–18, used under CC BY-NC-SA 4.0; OLS framing follows Stock and Watson, *Introduction to Econometrics*.